Assignment 3 — Decision Making & Control Flow

This assignment has two parts: a short multiple-choice check on Lesson 3 concepts, and two small coding problems that put if / else if / else and Boolean logic to work.

Part 1: Multiple Choice (__ / 8 points)

Check the box next to the best answer for each question. Each question is worth 5 points.

1. In a flowchart, what does a diamond symbol represent? *(5 pts)* ***[DAY 1]***

- ☐ A) A start or end point
- ☐ B) An action step
- ☐ C) A decision point
- ☐ D) A loop

2. Which of the following correctly checks whether x equals 10 (rather than assigning it)? *(5 pts)* ***[DAY 2]***

- ☐ A) if (x = 10)
- ☐ B) if (x == 10)
- ☐ C) if (x := 10)
- ☐ D) if (x =! 10)

3. What does the && (AND) operator require in order to evaluate to true? *(5 pts)* ***[DAY 1]***

- ☐ A) Only one side must be true
- ☐ B) Both sides must be true
- ☐ C) Neither side can be true
- ☐ D) Exactly one side must be false

4. Given int x = 45;, what does this code print? *(5 pts)* ***[DAY 2]***

```
if (x >= 60) { cout << "Pass"; } else { cout << "Fail"; }
```

- ☐ A) Pass
- ☐ B) Fail
- ☐ C) Nothing prints
- ☐ D) Compiler error

5. In an else-if chain, C++ checks the conditions: *(5 pts)* ***[DAY 2]***

- ☐ A) In random order
- ☐ B) From bottom to top
- ☐ C) Top to bottom, stopping at the first true one
- ☐ D) All conditions at the same time

6. Which value is treated as false in C++? *(5 pts)* ***[DAY 1]***

- ☐ A) Any negative number
- ☐ B) Only the number 0
- ☐ C) Any decimal number
- ☐ D) The word “false” only

7. What is the value of (7 > 3) || (2 == 5)? *(5 pts)* ***[DAY 1]***

- ☐ A) true
- ☐ B) false
- ☐ C) Compiler error
- ☐ D) Undefined

8. What happens if you accidentally write if (grade = 60) instead of if (grade == 60)? *(5 pts)* ***[DAY 2]***

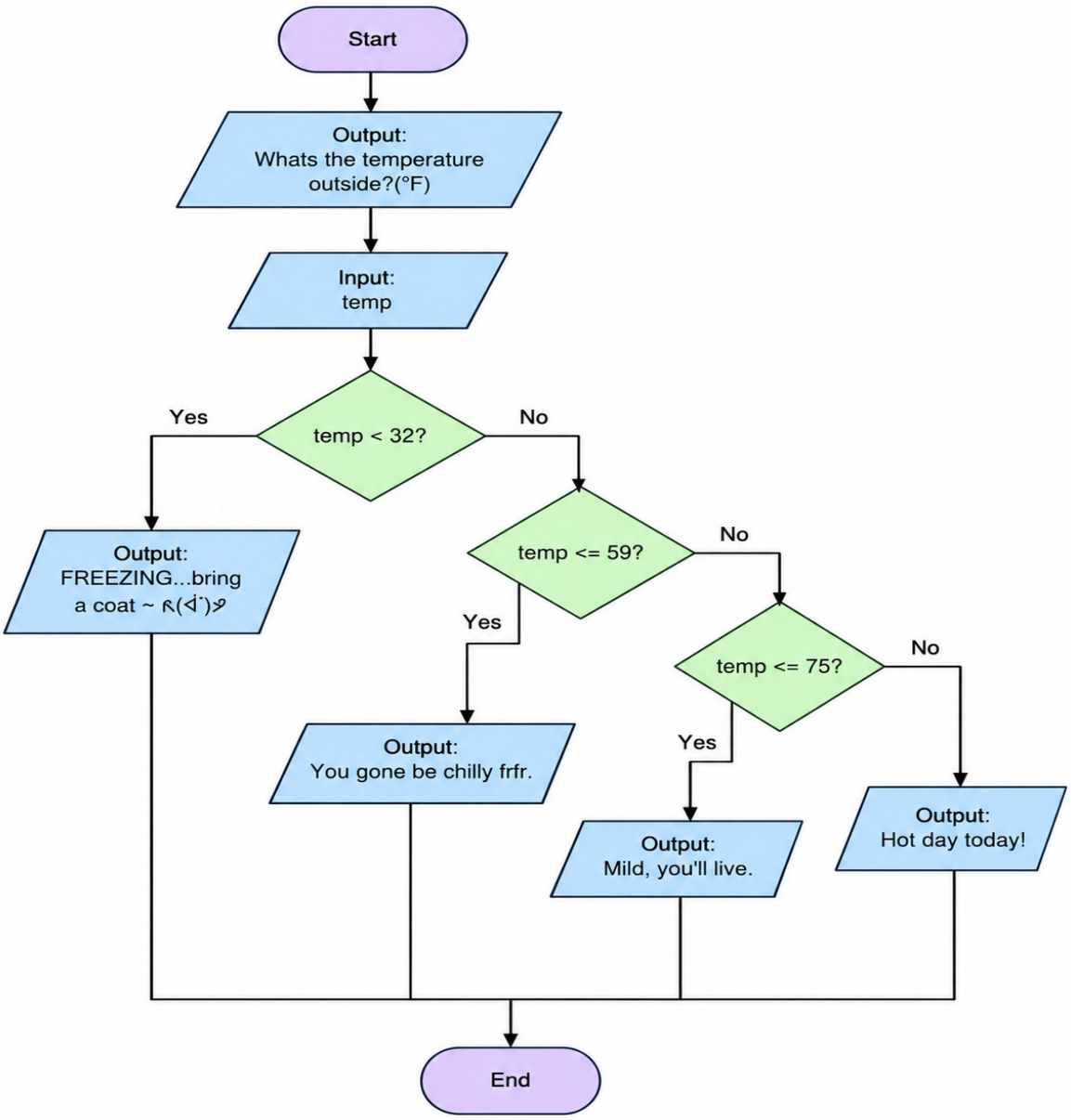
- ☐ A) The compiler catches the error immediately
- ☐ B) It assigns 60 to grade and the condition always evaluates as true
- ☐ C) It behaves identically to ==
- ☐ D) The program refuses to compile

Part 2: Coding Problems (__ / 22 points)

Since we won't be meeting Monday, these three problems cover the hands-on practice you would have gotten in that session — Boolean logic, relational operators, and if/else-if chains, each from a slightly different angle. Each problem is worth 20 points.

Problem 1: Temperature Classifier *(__ / 15 pts)* ***[DAY 2]***

(__ / 5 points) Draw the flow chart



(__ / 10 points) Write a C++ program called temp_check.cpp that classifies a Fahrenheit temperature.

- ☐ Ask the user to enter a temperature (as an integer) using cin<.
- ☐ Use an if / else if / else chain to classify it using these rules:

  Below 32 → “Freezing” 32–59 → “Cold” 60–75 → “Mild” 76 and above → “Hot”
- ☐ Print the classification clearly, e.g. “That's Mild.”

Example interaction:

```
Enter the temperature (F): 70
That's Mild.
```

Problem 2: Discount Eligibility *(__ / 7 pts)* ***[DAY 1 & 2]***

Write a C++ program called eligibility.cpp that checks whether someone qualifies for a discount.

- ☐ Ask the user for their age (an integer).
- ☐ Ask whether they're a member, entered as 1 for yes or 0 for no.
- ☐ A person qualifies if their age is 60 or older, OR if they're 18+ AND a member.
- ☐ This means combining && and || in one condition — use parentheses to make your intent clear, e.g. (age >= 18 && isMember == 1).
- ☐ Print whether they qualify or not, clearly.

Example interaction:

```
Enter your age: 20
Are you a member? (1 = yes, 0 = no): 1
You qualify for the discount.
```

Before You Submit
